

SQL SERVER INTERVIEW QUESTIONS AND ANSWERS

EASY LEVEL

1. What is the difference between a clustered and non-clustered index?

Clustered index sorts and stores data rows in the table; non-clustered index stores a separate structure with pointers.

2. Explain the purpose of SQL Server DMVs.

DMVs provide internal server state for diagnosing issues.

3. What are the common SQL Server services you check after installation?

SQL Server Engine, SQL Agent, Browser, SSIS, SSRS (if applicable).

4. How do you check SQL Server version and edition?

SELECT @@VERSION or use SQL Server Properties.

5. What is the difference between a full backup and a differential backup?

Full backup copies entire DB; differential copies changes since last full.

6. What is SQL Server Agent used for?

Scheduling and executing jobs.

7. Explain the use of SQL Profiler.

Captures and analyzes SQL events.

8. How do you kill a blocking session?

KILL .

9. What is a stored procedure?

A precompiled set of SQL statements.

10. What are the different types of joins?

Inner, Left, Right, Full, Cross.

MEDIUM LEVEL

1. How do you identify slow-running queries using DMVs?

Query sys.dm_exec_query_stats, sys.dm_exec_sql_text, sys.dm_exec_query_plan.

2. What steps do you follow when CPU is high?

Check top queries via DMVs, wait stats, plan cache, index issues.

3. Explain parameter sniffing and mitigation?

SQL reuses plan based on first parameter; fix via OPTION(RECOMPILE), optimize for unknown, or plan guides.

4. How do you check index fragmentation?

Use sys.dm_db_index_physical_stats.

5. What are missing indexes?

Indexes suggested by DMV sys.dm_db_missing_index*.

6. What are pre-checks before applying patches?

Backups, check compatibility, review release notes, test in QA.

7. Difference between in-place and side-by-side upgrade?

In-place upgrades existing instance; side-by-side installs new instance and migrates.

8. What is a maintenance plan?

Automated tasks like backups, integrity checks, index maintenance.

9. What is quorum in clustering?

Mechanism ensuring cluster consensus.

10. Difference between Always On AG and mirroring?

AG supports multiple DBs, replicas, read-only routing; mirroring is simpler and deprecated.

11. SQL vs Windows authentication?

SQL uses SQL logins; Windows uses AD identities.

12. How to audit failed logins?

Enable login auditing in server properties, check SQL error log.

13. How do you deploy SSIS packages?

Using ISDeploymentWizard or Integration Services Catalog.

14. Reasons for slow SSRS report?

Complex queries, dataset volume, inefficient expressions.

15. How to troubleshoot connectivity issues?

Check network, ping, port, SQL Browser, firewalls, protocols.

HARD LEVEL

1. End-to-end performance tuning?

Gather symptoms, analyze waits, top queries, indexing, plan analysis, hardware review.

2. How Query Optimizer works?

Generates execution plans via cost-based optimization.

3. What are wait stats?

Indicators of resource bottlenecks like PAGEIOLATCH (IO), CXPACKET (parallelism).

4. Cardinality estimation issues?

Incorrect row estimates due to outdated stats or parameter sniffing.

5. Parameter-sensitive plans?

Different param values need different plans; fix via recompile, hints, plan freezing.

6. SQL Server memory architecture?

Buffer Pool, Plan Cache, memory grants, NUMA nodes.

7. Build multi-node AG across DCs?

Create WSFC, configure quorum, create AG, setup endpoints, async replicas.

8. Planned vs unplanned failover?

Planned ensures no data loss; unplanned may have potential data loss.

9. Log send queue & redo queue issues?

Monitor DMVs; fix network latency, disk IO, bottlenecks.

10. DR strategy design?

Define RPO/RTO, choose AG/mirroring/log shipping, test failover.

11. Validate replication consistency?

Use table checksums or `sp_publication_validation`.

12. Dev fast but prod slow?

Compare plans, stats, indexes, resources, parameter sniffing.

13. CPU 100% but no queries?

Check system processes, spinlocks, outdated stats, background tasks.

14. Orphaned users?

Fix with `sp_change_users_login` or `ALTER USER WITH LOGIN`.

15. Deadlock detection?

Use system_health session or trace flag 1222.

16. TempDB contention?

Add multiple files, enable TF 1117/1118 (older versions), optimize queries.

17. Kerberos and SPN issues?

Set correct SPNs, use setspn tool, enable delegation.

18. SQL Server compliance hardening?

Encrypt data, enforce least privileges, auditing, TLS.

19. AD groups for access?

Assign server/DB roles to AD groups.

20. Optimize SSIS ETL?

Use fast load, avoid row-by-row, staging, partitioning.

21. SSRS scale-out deployment?

Multiple servers sharing report server DB.